

ALEX TAN MING WEI

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SUMMARY

Systems-oriented RTIS graduate with hands-on experience building real-time simulation systems, REST APIs, and containerised infrastructure tools.

PROJECTS

Distributed Task Scheduler (Aug 2025 ? Dec 2025)

- Designed a Python-based distributed task scheduler using Redis and Docker that reduced job processing time by 40%, handling 500+ concurrent tasks.
- Implemented RESTful APIs with FastAPI and deployed on AWS ECS using Terraform, achieving 99.9% uptime over a 3-month period.
- Automated CI/CD pipeline with GitHub Actions, cutting deployment time from 30 minutes to under 5 minutes.

Real-Time Physics Engine (Jan 2025 ? May 2025)

- Built a real-time application loop in C++ with Vulkan rendering that sustained 60 fps on 500+ dynamic objects simultaneously.
- Implemented spatial algorithms (AABB broad-phase, SAT narrow-phase) in C++, reducing collision computation time by 35%.
- Deployed cross-platform on Windows and Linux using CMake.

Linux Monitoring Dashboard (Mar 2024 ? Jun 2024)

- Developed a Prometheus + Grafana monitoring stack with Bash automation scripts that detected system anomalies 3x faster than manual log review.
- Containerised the stack with Docker Compose; integrated Kubernetes Helm charts for scaling across 10-node clusters.

EXPERIENCE

Systems Engineering Intern, Accenture Singapore (May 2025 ? Aug 2025)

- Automated infrastructure provisioning with Terraform and Ansible across AWS GCP environments, reducing manual setup time by 60%.
- Wrote Python scripts to parse and validate Kubernetes pod logs, enabling the team to resolve incidents 50% faster.

EDUCATION

DigiPen Institute of Technology Singapore

Bachelor of Science in Real-Time Interactive Simulation (RTIS), May 2026

Relevant courses: Operating Systems, Computer Networks, Distributed Systems, Computer Graphics, Algorithms & Data Structures

SKILLS

Languages: Python, C++, Go, Bash

Frameworks: FastAPI, Pytest, gRPC

Tools: Docker, Kubernetes, Terraform, Ansible, Prometheus, Grafana, GitHub Actions

Concepts: REST API, microservices, CI/CD, distributed systems, TCP/IP, DNS

Platforms: AWS, GCP, Linux